Snapshot 1 – Start Objective

The purpose of this project is to develop a software system capable of detecting copy-move forgery in biomedical and scientific images. Copy-move forgery occurs when a section of an image is duplicated and pasted elsewhere within the same image in order to manipulate scientific results.

The system will allow users to upload biomedical images through a web interface. The uploaded image will then be analyzed by a backend detection system that identifies potentially duplicated regions. The project is intended to help improve scientific integrity by assisting researchers, reviewers, and institutions in identifying manipulated figures.

This project will be developed incrementally throughout multiple project snapshots.

Objectives

Create a GitHub repository that will contain all project files, source code, documentation, and version history.

- Create GitHub repository
- Organize repository folder structure
- Add README file
- Create documentation folders

Objective 2 – Setup Frontend Framework

frontend structure for the web application.

- React.js
- HTML/CSS

Tasks

- Create React frontend application
- Configure project dependencies
- Add image upload interface

Objective 3 – Setup Backend Framework

Develop the backend API structure responsible for receiving uploaded images and communicating with the detection system.

- Python
- FastAPI
- Uvicorn

Tasks

- Create FastAPI backend server
- Configure backend dependencies
- Test server responses

Objective 4 – Define Initial Detection Workflow

Plan the workflow for how biomedical images will be processed.

Initial Workflow

1. User uploads image
2. Frontend sends image to backend
3. Backend processes request
4. Detection system analyzes image
5. Results returned to frontend

Objective 5 – Docker Environment Planning

Plan the containerized deployment structure for the project.

Planned Services

- Frontend container
- Backend container

Tasks

- Create initial docker-compose.yml file
- Define frontend service
- Define backend service
- Plan future ML service integration

Objective 6 - **Project documentation.**

Documents Started

- Software Design Document (SDD)
- Software Requirements Specification (SRS)
- Design Specification
- User Manual (README)
- Snapshot Objectives

Tasks

- Create LaTeX document templates
- Create version tables
- Begin introduction sections
- Document project purpose and architecture

4. Initial Dependencies and Libraries

Frontend Dependencies

- react
- react-dom

Backend Dependencies

- fastapi

Planned Future Dependencies

- torch
- pillow

5. Initial Team Task Delegation

Frontend Development

Responsibilities:

- User interface design
- Upload page implementation
- Displaying results

Backend Development

Responsibilities:

- API development
- Image upload handling
- Communication with detection system

Machine Learning / Detection

Responsibilities:

- Research copy-move forgery detection methods
- Implement image analysis
- Develop segmentation and detection logic

Documentation and Testing

Responsibilities:

- Maintain SRS and SDD
- Create TestRail reports
- Update Jira sprint tasks
- Maintain workflow diagrams

Snapshot 2 – Checkpoint 1

This snapshot focuses on expanding the project by implementing the frontend system, which serves as the user interface for uploading and interacting with biomedical image analysis.

At this stage, the project begins transitioning from planning and setup into actual system development.

Objective 1 – Frontend Implementation (New Major Feature)

Purpose

The purpose of this objective is to develop the frontend portion of the system that allows users to interact with the application through a web interface. This includes uploading biomedical images and preparing them to be sent to the backend for processing.

Frontend Framework

- HTML/CSS
- React

Tasks

- Create main user interface layout (home page and upload page)
- Implement image upload functionality
- Create basic navigation between UI sections

Expected Outcome for Snapshot 2

At the end of Snapshot 2, the frontend system should:

- Allow users to upload biomedical images
- Display a preview of selected images
- Have a structured UI ready for backend communication
- Be organized into reusable components
- Be prepared for integration with the backend system in future snapshots

Snapshot 3 – Checkpoint 2

The purpose of Snapshot 3 is to expand the Scientific Forgery Image Detection System by introducing image analysis functionality capable of identifying potentially fraudulent regions within uploaded images.

This snapshot focuses on implementing the detection logic that analyzes uploaded images and identifies areas that may contain copy move forgery.

Objective 1 – Fraudulent Region Detection System

Purpose

The purpose of this objective is to develop the initial detection functionality that identifies where potentially fraudulent or duplicated regions exist within images.

The system will begin analyzing uploaded images and returning basic detection results indicating suspicious areas that may have been manipulated through copy move forgery.

Detection System Technologies

- Python
- OpenCV
- NumPy
- Pillow

Tasks

- Research image duplication detection methods
- Implement basic image analysis functionality
- Detect repeated or duplicated image regions
- Generate basic highlighted output for suspicious regions
- Connect detection system to backend API
- Return detection results to frontend interface
- Prepare frontend to display fraudulent region results

Objective 2 – Backend and Frontend Integration

Purpose

Integrate the frontend image upload system with the backend detection functionality.

Tasks

- Connect React frontend to FastAPI backend
- Send uploaded image data to backend API
- Process backend responses
- Display returned analysis results in frontend interface
- Improve error handling for invalid uploads

Objective 3 – Detection Workflow Update

Updated Workflow

1. User uploads biomedical image
2. Frontend sends image to backend
3. Backend processes uploaded image
4. Detection system analyzes image for duplicated regions
5. Suspicious regions are identified
6. Detection results returned to frontend
7. User views highlighted suspicious regions

Objective 4 – Project Documentation Updates

Documents Updated

- Software Design Document (SDD)
- Software Requirements Specification (SRS)
- Design Specification
- Snapshot Objectives
- README/User Manual

Tasks

- Update workflow diagrams
- Document detection functionality
- Revise frontend/backend integration details
- Update version tables
- Document detection system architecture

Objective 5 – Testing and Validation

Tasks

- Test image upload functionality
- Test backend API communication
- Test fraudulent region detection responses
- Validate detection output formatting
- Document test results in TestRail

Snapshot 4 – Original Due Date

The purpose of Snapshot 4 is to finalize the Scientific Forgery Image Detection System by improving system stability, refining the user interface, and documenting future improvements that could be added beyond the current project scope.

This snapshot focuses on polishing the existing functionality developed in previous snapshots and preparing the project for final submission.

Objective 1 – Final System Improvements

Purpose

Improve the overall usability and organization of the Scientific Forgery Image Detection System.

Tasks

- Improve frontend layout and organization
- Refine image upload interface
- Improve display of detection results
- Clean up project folder structure
- Improve API response handling
- Fix minor frontend and backend issues

Objective 2 – Detection Result Refinements

Purpose

Improve the presentation of fraudulent region detection results returned to users.

Tasks

- Improve suspicious region highlighting
- Improve formatting of result messages
- Improve communication between frontend and backend
- Improve handling of invalid image uploads
- Improve detection result display structure

Objective 3 – Final Documentation Updates

Documents Updated

- Software Design Document (SDD)
- Software Requirements Specification (SRS)
- Design Specification
- Snapshot Objectives
- README/User Manual
- TestRail Reports

Tasks

- Finalize all project documents
- Update version tables
- Revise workflow descriptions
- Ensure documentation consistency across all files
- Finalize references and formatting